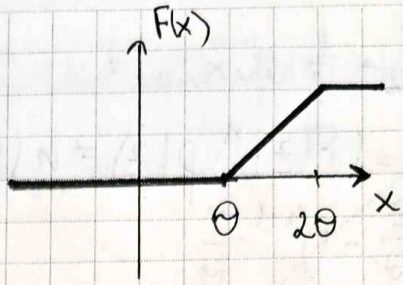
5. $\xi \sim R(\theta, 2\theta)$

$\vec{X}_n$ - выборка



Метод моментов

$$f(x,\theta) = \frac{1}{\theta}\{\theta, 2\theta\}$$

$$\alpha_1 = M\xi = \int_\theta^{2\theta} x \cdot \frac{1}{\theta} dx = \frac{x^2}{2\theta}\Big|_\theta^{2\theta} = \frac{3\theta}{2}$$

$$\alpha_2 = \frac{x^3}{3} \cdot \frac{1}{\theta}\Big|_\theta^{2\theta} = \frac{7\theta^2}{3}$$

$$D\xi = \frac{7\theta^2}{3} - \frac{9\theta^2}{4} = \frac{1}{12}\theta^2$$

$$\alpha_1 = \tilde{\alpha}_1 = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{X}$$

$$\tilde{\theta}_1 = \frac{2}{3}\bar{X}$$

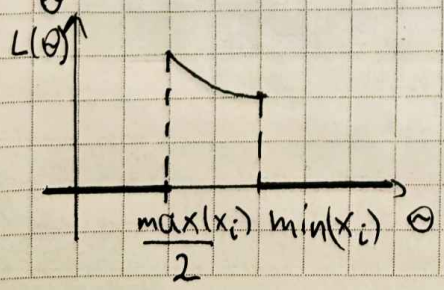
$$M\tilde{\theta}_1 = \frac{2}{3} M\bar{X} = \frac{2}{3} M\xi = \theta \text{ несмещ}$$

$$D\tilde{\theta}_1 = \frac{4}{9n} D\xi = \frac{1}{27n}\theta^2 \xrightarrow[n\to\infty]{} 0 \Rightarrow \text{состоятельная}$$

Метод максимального правдоподобия

$$L\theta = \prod_{i=1}^{n} P(x_i, \theta) = \frac{1}{\theta^n}\{\forall i \ \theta < x_i < 2\theta\} =$$

$$= \frac{1}{\theta^n}\{\min(x_i) > 0, \max(x_i) < 2\theta\}$$



$$\Rightarrow \tilde{\theta}_2 = \frac{x_{max}}{2}$$

$\varphi(z) = (F(z))^n \qquad \varphi(z) = n(F(z))^{n-1} p(z) = n\left(\frac{z}{\theta} - 1\right)^{n-1} \cdot \frac{1}{\theta}$

$$M\tilde{\theta}_2 = \frac{1}{2} M x_{max} = \frac{1}{2} \int_{\theta}^{2\theta} z n \left(\frac{z}{\theta} - 1\right)^{n-1} \frac{1}{\theta} dz =$$

$$= \frac{\theta}{2} \int_1^2 n t (t-1)^{n-1} dt = \left\{ \begin{array}{ll} t = u & dt = du \\ n(t-1)^{n-1} dt = dv & \\ & v = (t-1)^n \end{array} \right\} =$$

$$= \frac{\theta}{2} \left[t(t-1)^n \Big|_1^2 - \int_1^2 (t-1)^n dt \right] = \frac{\theta}{2} \left[2 - \frac{(t-1)^{n+1}}{n+1} \Big|_1^2 \right] =$$

$$= \frac{\theta}{2} \left[2 - \frac{1}{n+1} \right] = \frac{\theta}{2} \left[\frac{2n+2-1}{n+1} \right] = \frac{(2n+1)\theta}{2n+2} \text{ смещ}$$

$$\tilde{\theta}_2^1 = \frac{2n+2}{2n+1} \tilde{\theta}_2 \text{ несмещ}$$

$$M\tilde{\theta}_2^2 = \int_{-\infty}^{\infty} z^2 \varphi(z) dz = \frac{1}{2} \int_{\theta}^{2\theta} n z^2 \left(\frac{z}{\theta} - 1\right)^{n-1} \frac{1}{\theta} dz =$$

$$= \frac{1}{4} \int_{\theta}^{2\theta} n \frac{z^2}{\theta^2} \theta \left(\frac{z}{\theta} - 1\right)^{n-1} \frac{1}{\theta} dz = \frac{\theta^2}{4} \int_1^2 n t^2 (t-1)^{n-1} dt =$$

$$= \frac{\theta^2}{4} \left[t^2 (t-1)^n \Big|_1^2 - 2 \int_1^2 t (t-1)^n dt \right] = \frac{\theta^2}{4} \left[4 - 2 \left[t - \frac{(t-1)^{n+1}}{n+1} \Big|_1^2 - \right.\right.$$

$$\left.\left. - \int_1^2 \frac{(t-1)^{n+1}}{n+1} dt \right] \right] = \theta^2 - \frac{\theta^2}{2} \left[\frac{2}{n+1} - \frac{(t-1)^{n+2}}{(n+1)(n+2)} \Big|_1^2 \right] =$$

$$= \theta^2 - \frac{\theta^2}{2} \left[\frac{2(n+2)-1}{(n+1)(n+2)} \right] = \theta^2 - \frac{\theta^2}{2} \frac{2n+3}{(n+1)(n+2)} =$$

$$= \frac{\theta^2 2(n^2+3n+2) - \theta^2(2n+3)}{2(n+1)(n+2)} = \frac{2n^2\theta^2 + 4\theta^2 n + \theta^2}{2(n+1)(n+2)} =$$

$$= \frac{\theta^2}{2} \cdot \frac{2n^2+4n+1}{(n+1)(n+2)}$$

$$D\tilde{\theta}_2 = \left(\frac{2n^2+4n+1}{2(n+1)(n+2)} - \frac{(2n+1)^2}{4(n+1)^2}\right)\theta^2 =$$

$$= \frac{2(n+1)(2n^2+4n+1) - (2n+1)^2(n+2)}{4(n+1)^2(n+2)}\theta^2 = \frac{\theta^2 n}{4(n+1)^2(n+2)}$$

$$D\tilde{\theta}_2' = \left(\frac{2n+2}{2n+1}\right)^2 \cdot \frac{n\theta^2}{4(n+1)^2(n+2)} = \frac{n}{(2n+1)^2(n+2)}\theta^2 \xrightarrow[n\to\infty]{} 0$$

=> состоятельная

$$D\tilde{\theta}_1 = \frac{\theta^2}{27n} \qquad D\tilde{\theta}_2' = \frac{n\theta^2}{(2n+1)^2(n+2)}$$

$D\tilde{\theta}_2' < D\tilde{\theta}_1 \quad \forall n \geq 3$

=> $\tilde{\theta}_2'$ эффективнее $\tilde{\theta}_1$

d) Точный интервал $\quad F(t) = \begin{cases} 0, & t < \theta \\ \frac{t}{\theta} - 1, & \theta \leq t \leq 2\theta \\ 1, & t > 2\theta \end{cases}$

$$f(\vec{x}_n, \theta) = \frac{\max(x_1 \dots x_n)}{\theta} - 1$$

$$q(t) = P(f < t) = P(\max(x_1, \dots, x_n) < \theta t + \theta) =$$

$$= P(x_1 < \theta t + \theta, \dots, x_n < \theta t + \theta) = P((x_1 < \theta t + \theta)^n) =$$

$$= (F(\theta t + \theta))^n$$

$$q(t) = \begin{cases} 0, & t < 0 \\ t^n, & 0 \leq t \leq 1 \\ 1, & t > 1 \end{cases}$$

$$q(x_{\frac{\alpha}{2}}) = \frac{\alpha}{2} \qquad q(x_{\beta+\frac{\alpha}{2}}) = \beta + \frac{\alpha}{2}$$

$$t_1 = x_{\frac{\alpha}{2}} = \sqrt[n]{\frac{\alpha}{2}} = \sqrt[n]{\frac{1-\beta}{2}}$$

$$t_2 = x_{\frac{1+\beta}{2}} = \sqrt[n]{\frac{1+\beta}{2}}$$

$$t_1 < \frac{\max x_i}{\theta} - 1 < t_2$$

$$\frac{\max x_i}{\left(\frac{1+\beta}{2}\right)^{1/n+1}} < \theta < \frac{\max x_i}{\left(\frac{1-\beta}{2}\right)^{1/n+1}}$$

е) асимптотический

ОММ: $\tilde{\theta} = \frac{2}{3}\bar{x} = \frac{2}{3}\tilde{\alpha}_1 = g(\tilde{\alpha}_1)$

$$\frac{\sqrt{n}\,(g(\tilde{\alpha}_1) - g(\alpha_1))}{\sqrt{\nabla^T g K \nabla g}} \rightsquigarrow N(0,1)$$

$\nabla g = \frac{2}{3} \quad K_{11} = \alpha_2 - \alpha_1^2$

$$\frac{\sqrt{n}(\tilde{\theta}_1 - \theta)}{\frac{2}{3}\sqrt{\tilde{\alpha}_2 - \tilde{\alpha}_1^2}} \rightsquigarrow N(0,1)$$

$$\frac{n(\tilde{\theta}_1 - \theta)}{\frac{2}{3} S\sqrt{n-1}} \rightsquigarrow N(0,1)$$

$$P\left(t_1 < \frac{3n(\tilde{\theta}_1 - \theta)}{2S\sqrt{n-1}} < t_2\right) = \beta$$

$$\tilde{\theta}_1 - \frac{t_2 \cdot 2S\sqrt{n-1}}{3n} < \theta < \tilde{\theta}_1 - \frac{t_1 \cdot 2S\sqrt{n-1}}{3n}$$